Niobium

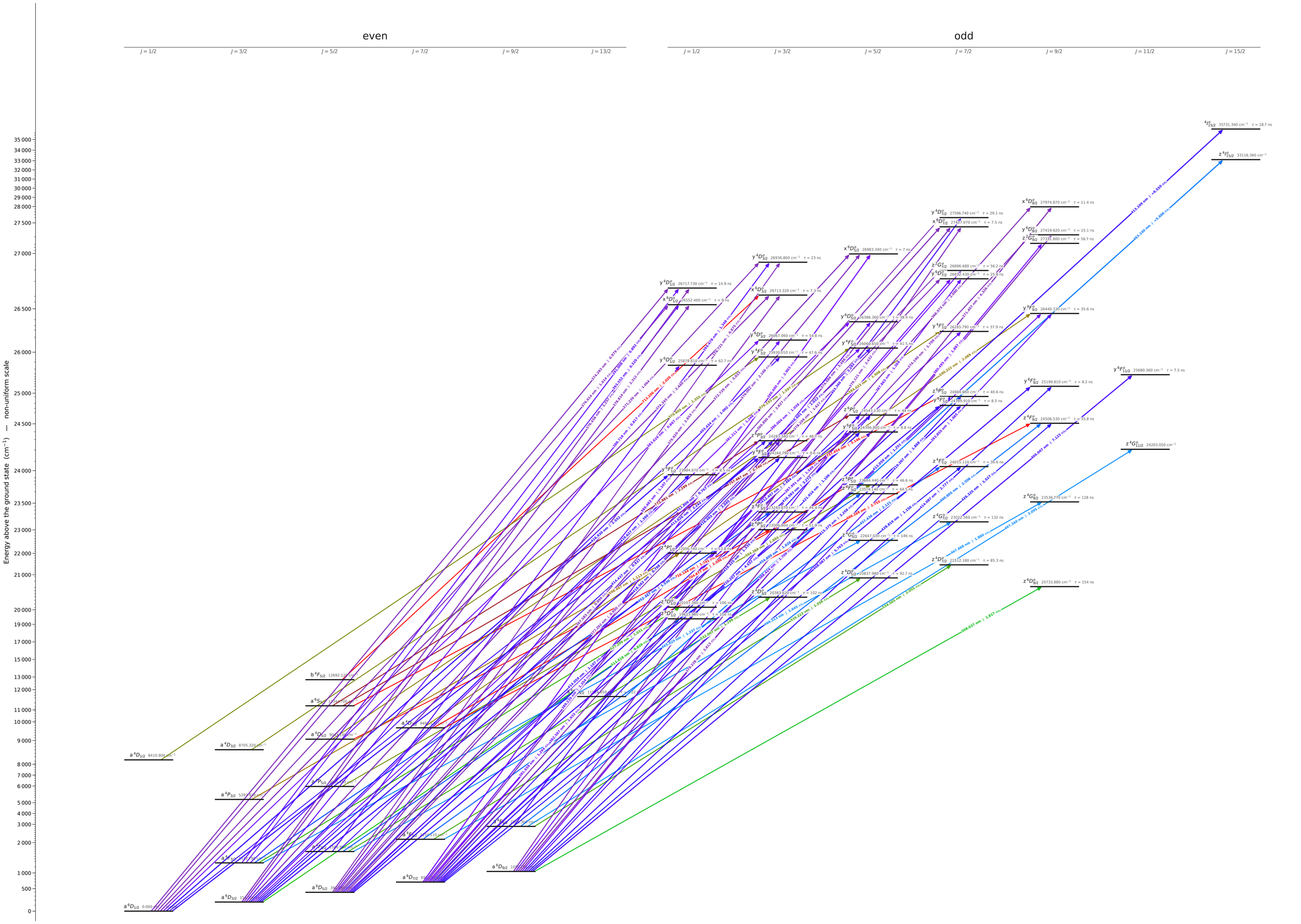
Nb (Nb I). Strongest classified transitions below 2 µm (NIST): 65 levels, 90 lines.

Nb^+ ionisation limit 54513.80 cm^{-1} = 6.75885 eV (183.440 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	|⟨J‖er‖J'⟩| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a $^4H_{13/2}$ – $^4I^o_{15/2}$	413.1091	412.9926	725.69801(10)	≈6.550	7.707e+07	8.511	100	second proposed laser-cooling line from metastable a4H13/...
a $^4H_{13/2}$ – z $^4I^o_{15/2}$	463.1405	463.0108	647.30348(10)	≈5.696	4.136e+07	–	–	proposed laser-cooling line from metastable a4H13/2 at 46...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{93}Nb hyperfine structure (I = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a $^6D_{1/2}$	1868.2184(18)	–	F=4: -5137.60 F=5: +4203.49
a $^6D_{3/2}$	852.5428(21)	-64.57(4)	F=3: -7063.07 F=4: -3609.85 F=5: +666.31 F=6: +5738.52
a $^6D_{5/2}$	719.4759(18)	-47.60(4)	F=2: -9914.61 F=3: -7741.90 F=4: -4850.51 F=5: -1245.20 F=6: +3068.09 F=7: +8082.20
a $^6D_{7/2}$	690.1978(16)	20.48(8)	F=1: -13276.92 F=2: -11899.45 F=3: -9832.64 F=4: -7075.75 F=5: -3627.80 F=6: +512.41 F=7: +5346.35 F=8: +10875.73
a $^6D_{9/2}$	691.6141(22)	132.75(12)	F=0: -17056.60 F=1: -16372.36 F=2: -15002.97 F=3: -12946.56 F=4: -10200.39 F=5: -6760.75 F=6: -2623.05 F=7: +2218.24 F=8: +7769.60 F=9: +14038.37
a $^4F_{3/2}$	644.2(5)	32.80(13)	F=3: -5299.29 F=4: -2744.51 F=5: +469.45 F=6: +4356.28
a $^4F_{5/2}$	372.4853(17)	33.34(4)	F=2: -5106.39 F=3: -3998.94 F=4: -2518.44 F=5: -661.57 F=6: +1575.84 F=7: +4198.80
a $^4F_{7/2}$	292.2022(16)	44.93(5)	F=1: -5604.30 F=2: -5026.31 F=3: -4158.00 F=4: -2997.75 F=5: -1543.42 F=6: +207.65 F=7: +2258.68 F=8: +4613.42
a $^4F_{9/2}$	269.6320(26)	64.31(13)	F=0: -6643.91 F=1: -6377.85 F=2: -5845.29 F=3: -5045.33 F=4: -3976.62 F=5: -2637.40 F=6: -1025.41 F=7: +862.01 F=8: +3028.00 F=9: +5476.13
a $^4P_{3/2}$	497.7569(16)	59.902(14)	F=3: -4079.04 F=4: -2127.95 F=5: +348.36 F=6: +3374.84
a $^4P_{5/2}$	343.1670(5)	-80.454(9)	F=2: -4755.42 F=3: -3701.78 F=4: -2306.32 F=5: -577.08 F=6: +1475.89 F=7: +3840.51
a $^4D_{1/2}$	1997.0(11)	–	F=4: -5491.75 F=5: +4493.25
a $^4D_{3/2}$	-143.322(22)	-10.76(18)	F=3: +1177.48 F=4: +611.36 F=5: -103.01 F=6: -970.12
a $^4D_{5/2}$	-407.861(4)	35.06(8)	F=2: +5624.15 F=3: +4390.05 F=4: +2748.68 F=5: +703.53 F=6: -1741.00 F=7: -4579.67
a $^4D_{7/2}$	-477.037(4)	126.90(8)	F=1: +9241.13 F=2: +8268.93 F=3: +6814.40 F=4: +4882.08 F=5: +2478.01 F=6: -390.26 F=7: -3713.66 F=8: -7481.61
a $^6S_{5/2}$	-639.379(8)	-0.06(9)	F=2: +8791.43 F=3: +6873.31 F=4: +4315.81 F=5: +1118.93 F=6: -2717.35 F=7: -7193.02
a $^4H_{13/2}$	662(20)	–	F=2: -22342.50 F=3: -20356.50 F=4: -17708.50 F=5: -14398.50 F=6: -10426.50 F=7: -5792.50 F=8: -496.50 F=9: +5461.50 F=10: +12081.50 F=11: +19363.50
b $^4F_{5/2}$	338.500	–	F=2: -4654.38 F=3: -3638.88 F=4: -2284.88 F=5: -592.38 F=6: +1438.62 F=7: +3808.12
z $^6D^o_{1/2}$	1713.50(20)	–	F=4: -4712.12 F=5: +3855.38
z $^4D^o_{1/2}$	-794.7(5)	–	F=4: +2185.43 F=5: -1788.08
z $^4D^o_{3/2}$	597.4(3)	–	F=3: -4928.55 F=4: -2538.95 F=5: +448.05 F=6: +4032.45
z $^6D^o_{3/2}$	733.9(4)	–	F=0: -18164.03 F=1: -17430.12 F=2: -15962.33 F=3: -13760.62 F=4: -10825.02 F=5: -7155.52 F=6: -2752.12 F=7: +2385.18 F=8: +8256.38 F=9: +14861.48
z $^4D^o_{5/2}$	809.80(10)	–	F=2: -11134.75 F=3: -8705.35 F=4: -5466.15 F=5: -1417.15 F=6: +3441.65 F=7: +9110.25
z $^4D^o_{7/2}$	872.600	–	F=1: -16797.55 F=2: -15052.35 F=3: -12434.55 F=4: -8944.15 F=5: -4581.15 F=6: +654.45 F=7: +6762.65 F=8: +13743.45
z $^4P^o_{1/2}$	-268.0(20)	–	F=4: +737.00 F=5: -603.00
z $^4G^o_{5/2}$	-78.50(20)	–	F=2: +1079.38 F=3: +843.88 F=4: +529.88 F=5: +137.38 F=6: -333.62 F=7: -883.12
z $^4P^o_{3/2}$	-138.9(3)	–	F=3: +1145.92 F=4: +590.33 F=5: -104.17 F=6: -937.58
z $^4G^o_{7/2}$	430.30(20)	–	F=1: -8283.27 F=2: -7422.68 F=3: -6131.77 F=4: -4410.57 F=5: -2259.07 F=6: +322.73 F=7: +3334.82 F=8: +6777.23
z $^4F^o_{3/2}$	-16.6(3)	–	F=3: +136.95 F=4: +70.55 F=5: -12.45 F=6: -112.05
z $^4G^o_{9/2}$	632.60(10)	–	F=0: -15656.85 F=1: -15024.25 F=2: -13759.05 F=3: -11861.25 F=4: -9330.85 F=5: -6167.85 F=6: -2372.25 F=7: +2055.95 F=8: +7116.75 F=9: +12810.15
z $^4F^o_{5/2}$	514.70(20)	–	F=2: -7077.12 F=3: -5533.02 F=4: -3474.22 F=5: -900.73 F=6: +2187.47 F=7: +5790.38
z $^4P^o_{5/2}$	103.5(4)	–	F=2: -1423.12 F=3: -1112.62 F=4: -698.62 F=5: -181.12 F=6: +439.88 F=7: +1164.38
y $^6F^o_{1/2}$	1474.3(6)	–	F=4: -4054.32 F=5: +3317.18
z $^4F^o_{7/2}$	592.20(10)	–	F=1: -11399.85 F=2: -10215.45 F=3: -8438.85 F=4: -6070.05 F=5: -3109.05 F=6: +444.15 F=7: +4589.55 F=8: +9327.15
y $^6F^o_{3/2}$	266.9(5)	–	F=3: -2201.93 F=4: -1134.33 F=5: +200.18 F=6: +1801.58
z $^4G^o_{11/2}$	752.30(10)	–	F=1: -22004.78 F=2: -20500.17 F=3: -18243.28 F=4: -15234.08 F=5: -11472.58 F=6: -6958.77 F=7: -1692.67 F=8: +4325.73 F=9: +11096.42 F=10: +18619.42
z $^6P^o_{3/2}$	90.9(4)	–	F=3: -749.92 F=4: -386.32 F=5: +68.17 F=6: +613.58
y $^6F^o_{5/2}$	227.50(10)	–	F=2: -3128.12 F=3: -2445.62 F=4: -1535.62 F=5: -398.12 F=6: +966.88 F=7: +2559.38
z $^4F^o_{9/2}$	593.8(5)	–	F=0: -14696.55 F=1: -14102.75 F=2: -12915.15 F=3: -11133.75 F=4: -8758.55 F=5: -5789.55 F=6: -2226.75 F=7: +1929.85 F=8: +6680.25 F=9: +12024.45
z $^6P^o_{5/2}$	136.80(20)	–	F=2: -1881.00 F=3: -1470.60 F=4: -923.40 F=5: -239.40 F=6: +581.40 F=7: +1539.00
y $^6F^o_{7/2}$	138.6(3)	–	F=1: -2668.05 F=2: -2390.85 F=3: -1975.05 F=4: -1420.65 F=5: -727.65 F=6: +103.95 F=7: +1074.15 F=8: +2182.95
z $^6P^o_{7/2}$	194.7(3)	–	F=1: -3747.97 F=2: -3358.57 F=3: -2774.47 F=4: -1995.67 F=5: -1022.17 F=6: +146.03 F=7: +1508.92 F=8: +3066.53
y $^6F^o_{9/2}$	112.4(4)	–	F=0: -2781.90 F=1: -2669.50 F=2: -2444.70 F=3: -2107.50 F=4: -1657.90 F=5: -1095.90 F=6: -421.50 F=7: +365.30 F=8: +1264.50 F=9: +2276.10
y $^6F^o_{11/2}$	44.90(10)	–	F=1: -1313.33 F=2: -1223.53 F=3: -1088.83 F=4: -909.23 F=5: -684.73 F=6: -415.32 F=7: -101.03 F=8: +258.18 F=9: +662.27 F=10: +1111.28
y $^6D^o_{1/2}$	1537.900	–	F=4: -4229.23 F=5: +3460.28
y $^4F^o_{3/2}$	559.7(4)	–	F=3: -4617.52 F=4: -2378.72 F=5: +419.77 F=6: +3777.97
y $^4F^o_{5/2}$	265.9(9)	–	F=2: -3656.12 F=3: -2858.43 F=4: -1794.83 F=5: -465.32 F=6: +1130.08 F=7: +2991.38
y $^6D^o_{3/2}$	933(13)	–	F=3: -7698.90 F=4: -3966.10 F=5: +699.90 F=6: +6299.10
y $^4F^o_{7/2}$	297.6(7)	–	F=1: -5728.80 F=2: -5133.60 F=3: -4240.80 F=4: -3050.40 F=5: -1562.40 F=6: +223.20 F=7: +2306.40 F=8: +4687.20
y $^6D^o_{5/2}$	804(6)	–	F=2: -11056.38 F=3: -8644.08 F=4: -5427.68 F=5: -1407.17 F=6: +3417.43 F=7: +9046.12
y $^4F^o_{9/2}$	337.4(8)	–	F=0: -8350.65 F=1: -8013.25 F=2: -7338.45 F=3: -6326.25 F=4: -4976.65 F=5: -3289.65 F=6: -1265.25 F=7: +1096.55 F=8: +3795.75 F=9: +6832.35
x $^6D^o_{1/2}$	430.300	–	F=4: -1183.33 F=5: +968.17
x $^6D^o_{3/2}$	104(5)	–	F=3: -854.70 F=4: -440.30 F=5: +77.70 F=6: +699.30
y $^4D^o_{1/2}$	506(4)	–	F=4: -1391.50 F=5: +1138.50
y $^6D^o_{7/2}$	378.400	–	F=1: -7284.20 F=2: -6527.40 F=3: -5392.20 F=4: -3878.60 F=5: -1986.60 F=6: +283.80 F=7: +2932.60 F=8: +5959.80
z $^2G^o_{7/2}$	181.300	-21.000	F=1: -3499.65 F=2: -3134.05 F=3: -2586.28 F=4: -1857.08 F=5: -947.45 F=6: +141.35 F=7: +1407.83 F=8: +2850.22
y $^4D^o_{3/2}$	39.9(6)	–	F=3: -329.18 F=4: -169.57 F=5: +29.93 F=6: +269.32
x $^6D^o_{5/2}$	196(4)	–	F=2: -2700.50 F=3: -2111.30 F=4: -1325.70 F=5: -343.70 F=6: +834.70 F=7: +2209.50
y $^6D^o_{9/2}$	405.700	47.000	F=0: -10019.53 F=1: -9616.44 F=2: -8809.94 F=3: -7599.37 F=4: -5983.75 F=5: -3961.78 F=6: -1531.82 F=7: +1308.08 F=8: +4560.21 F=9: +8227.17
x $^6D^o_{7/2}$	384.7(22)	–	F=1: -7405.48 F=2: -6636.07 F=3: -5481.98 F=4: -3943.18 F=5: -2019.67 F=6: +288.52 F=7: +2981.43 F=8: +6059.02
y $^4D^o_{7/2}$	-144.0(4)	–	F=1: +2772.00 F=2: +2484.00 F=3: +2052.00 F=4: +1476.00 F=5: +756.00 F=6: -108.00 F=7: -1116.00 F=8: -2268.00
x $^6D^o_{9/2}$	525.10(10)	–	F=0: -12996.23 F=1: -12471.12 F=2: -11420.92 F=3: -9845.62 F=4: -7745.23 F=5: -5119.73 F=6: -1969.12 F=7: +1706.58 F=8: +5907.38 F=9: +10633.27
z $^4I^o_{15/2}$	627(20)	–	F=3: -23986.58 F=4: -21478.17 F=5: -18342.67 F=6: -14580.08 F=7: -10190.38 F=8: -5173.57 F=9: +470.32 F=10: +6741.32 F=11: +13639.42 F=12: +21164.62
$^4I^o_{15/2}$	299(20)	–	F=3: -11429.10 F=4: -10233.90 F=5: -8739.90 F=6: -6947.10 F=7: -4855.50 F=8: -2465.10 F=9: +224.10 F=10: +3212.10 F=11: +6498.90 F=12: +10084.50

Measured A, B constants; sources in the data file.